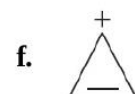
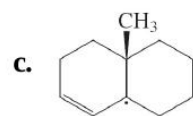
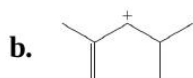
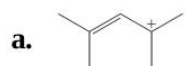


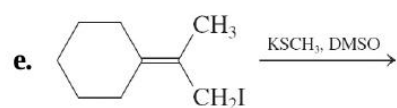
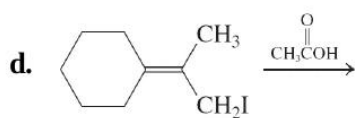
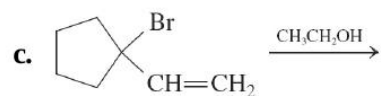
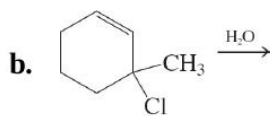
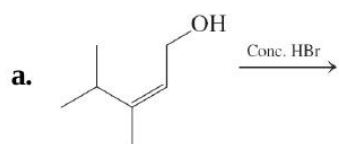
32. Draw all resonance forms and a representation of the appropriate resonance hybrid for each of the following species.



33. For each species in [Problem 32](#), indicate the resonance form that is the major contributor to the resonance hybrid. Explain your choices.

- 34.** Illustrate by means of appropriate structures (including all relevant resonance forms) the initial species formed by **(a)** breaking the weakest C—H bond in 1-butene; **(b)** treating 4-methylcyclohexene with a powerful base (e.g., butyllithium-TMEDA); **(c)** heating a solution of 3-chloro-1-methylcyclopentene in aqueous ethanol.

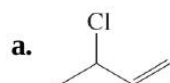
36. Give the major product(s) of each of the following reactions.



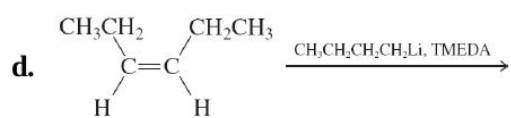
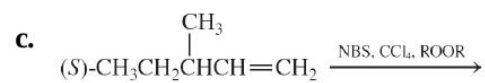
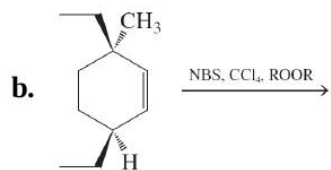
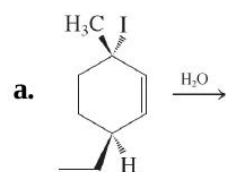
37. Formulate detailed mechanisms for the reactions in [Problem 36](#), parts (a), (c), (e), and (f).

38. Rank primary, secondary, tertiary, and (primary) allylic chlorides in approximate order of **(a)** decreasing $\text{S}_{\text{N}}1$ reactivity; **(b)** decreasing $\text{S}_{\text{N}}2$ reactivity.

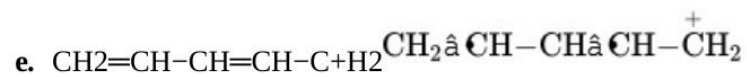
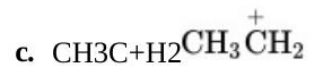
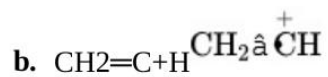
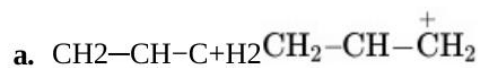
39. Rank the following six molecules in approximate order of decreasing $\text{S}_{\text{N}}1$ reactivity and decreasing $\text{S}_{\text{N}}2$ reactivity.



41. Give the major product(s) of each of the following reactions.



54. Arrange the following carbocations in order of decreasing stability.
Draw all possible resonance forms for each of them.



60. Give the product(s) of each of the following reactions.

- a. 3-Chloro-1-propene (allyl chloride) + NaOCH₃ ^{NaOCH₃}
- b. *cis*-2-Butene + NBS, peroxide (ROOR)
- c. 3-Bromocyclopentene + LDA
- d. *trans,trans*-2,4-Hexadiene + HCl ^{HCl}
- e. *trans,trans*-2,4-Hexadiene + Br₂, H₂O ^{Br₂, H₂O}
- f. 1,3-Cyclohexadiene + methyl propenoate (methyl acrylate)
- g. 1,2-Dimethylenecyclohexane + methyl propenoate (methyl acrylate)